

Bus Reservation & Fleet Scheduling System

Design and Implementation of a Relational Database Management System

Presented by

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Problem Identification



Seat Double Allocation

Manual reservation sheets lead to accidental double booking of the same seat on single trips.



Crew Duty Overlaps

Drivers and conductors are silently assigned to overlapping trip schedules without validation.



Uncentralized Operations

Fragmented tracking for schedules, payments, cancellations, and intermediate route stops.

Project Scope

System Boundaries & Deliverables

- ✓ **Relational Model:** Complete 11-table Normalized Database up to 3NF.
- ✓ **Data Integrity:** Strict primary keys, foreign keys, and trigger constraints.
- ✓ **Analytics Engine:** Reporting views for occupancy rates and route revenue.
- ✓ **User Interface:** Lightweight Front-End Application for core transactions.

Fleet & Inventory Management

Buses, physical seats, operational routes, and intermediate stops.

Passenger Booking Lifecycle

Reservations, seat selection, cancellations, and payment processing.

Crew Assignment & Operations

Driver and conductor duty schedules mapped directly to trip times.

Project Objectives

1

Normalize Schema (3NF)

Eliminate data redundancy, update anomalies, and partial dependencies across tables.

2

Prevent Booking Conflicts

Enforce dynamic seat allocation rules to restrict duplicate seat reservations on trips.

3

Automate Crew Rostering

Validate shift times to guarantee zero driver and conductor schedule duty overlaps.

4

Deliver Analytical Views

Provide simplified SQL reporting views for revenue per route and trip occupancy statistics.

Functional Requirements



1. Fleet & Routes

- Manage bus registrations and capacity
- Configure origin, destination & distance
- Define intermediate stop sequences



2. Trips & Crew

- Schedule departure and arrival times
- Assign driver and conductor duties
- Validate duty conflict restrictions



3. Bookings & Fares

- Real-time seat availability check
- Single payment record per booking
- Process booking cancellations

PROJECT REVIEW 1 SUMMARY

Thank You

With scope and requirements defined, next steps focus on ER modeling, schema normalization (3NF), DDL creation, and SQL constraint enforcement.